CIDR=Classless Inter Domain Routing is a method to allocating IP addresses.
CIDR allows routers to organize IP Addresses for multiple subnets or network and for devices into that Network.

eg.= 192.167.0.0/16

here /16 is Network identifier, so first 16 bits are Network portion whereas remaining bits are host portion.

An IPv4 always has 32 bits and divided into 4 Octets in "0" and "1"

In Above IP address 192 and 167 are On bits(1) and 0.0 are off bits(0) and /16 is network portion which are on bits

192.167.0.0=11111111.11111111.00000000.00000000 = 255.255.0.0

for hosts bits=2^32-n   n=number after "/"

so from above Ip host bits= 2^32-16 = 2^16 = 65,536

so 65,536 total usable addresses can be available in which first Ip 192.167.0.0 is network address

and last Ip 192.167.255.255 is broadcast address.so total usable IP are "65,534".

/8,/12,/16 are used for private network so for range we have to divide it into smaller subnet by borrowing bits

eg.= 192.168.0.0/20

20 bits are on and remaining are off

11111111.11111111.11110000.00000000 = 255.255.240.0 = unstable octet is 11110000

binary to decimal of 11110000 = 240

if octet contain all "1" then it is stable and denoted as "255"

if octet contain all "0" then it is stable and denoted as "0"

if octet contain all bits are "0" and "1" then it is Unstable and we have to convert it into decimal form

host bits= 2^32-n = 2^32-20 = 2^12 = 4096 total addresses in which one is network and one is broadcast so 4094 are usable Ip.

subnet range= 256-240 = 16

192.168.0.1----192.168.15.254
192.168.16.1----192.168.31.254
192.168.32.1----192.168.47.254
192.168.48.1----192.168.64.254 and so on upto 192.168.255.254